Cost and Safety of Label Acquisition Strategies for MALDI-TOF Antimicrobial Resistance Prediction

Muhammad Dyen Asif
Department of Computer Science
National University of Computer and Emerging Sciences
Islamabad, Pakistan
i240608@isb.nu.edu.pk

Hira Mastoor
Department of Computer Science
National University of Computer and Emerging Sciences
Islamabad, Pakistan
hira.mastoor@nu.edu.pk

Maryam Shahbaz
Department of Computer Science
National University of Computer and Emerging Sciences
Islamabad, Pakistan
maryam.shahbaz@nu.edu.pk

***Abstract*—Machine learning (ML) applied to matrix-assisted laser desorption ionization time-of-flight (MALDI-TOF) mass spectra predicts antimicrobial resistance (AMR) in hours rather than the 1 to 3 days that culture based susceptibility testing takes. A deployed model decays as the local population of isolates changes, so it has to be retrained, and every label is a laboratory susceptibility test that someone pays for. Active learning (AL) is the standard answer to an expensive label, yet a 2026 review of 115 ML studies on clinical MALDI-TOF spectra identified none in which the cost of acquiring labels was a variable. We ran 1540 retraining simulations on the Database of Resistance Information on Antimicrobials and MALDI-TOF Mass Spectra (DRIAMS), covering 3 species and antibiotic pairs, 4 hospitals, 5 budgets and 10 seeds per configuration. Models were fitted on 2015 and 2016, bought labels during 2017 in 4 rounds, and were tested on 2018. Across 135 comparisons no uncertainty, diversity or hybrid rule beat random selection after correction for multiple testing, and none did consistently once the batches were redrawn. Little accuracy was available, since the interval between never retraining and retraining on every arriving isolate spans 0.044 to 0.051 of the area under the receiver operating characteristic curve (AUROC). A budget buys cost rather than accuracy, and 400 tests (19% to 31% of a year of isolates) recovered 50% to 77% of that interval. Uncertainty sampling, the weakest rule by AUROC, cut the resistant isolates reported susceptible during the pool year by 33% to 73%.**

***Keywords*—antimicrobial resistance, active learning, MALDI-TOF mass spectrometry, model maintenance, explainable machine learning.**

I. Introduction

Antimicrobial resistance is among the largest threats to public health, and an estimated 1.27 million deaths were attributable to bacterial AMR in 2019 [1]. Treatment depends on knowing which drugs still work, which in routine practice means culture based AST, and that takes up to 72 hours. MALDI-TOF mass spectrometry is already present in most clinical microbiology laboratories for species identification, and the spectrum acquired for that purpose can be reused to predict resistance with ML, which reduces the delay from days to hours [2].

Such a model does not stay accurate on its own. The population of isolates in a hospital changes, the instrument is serviced or replaced, and the resistance mechanisms in circulation shift. A model trained here on isolates from 2015 and 2016 reached an AUROC of 0.881 under cross validation inside those years for *Escherichia coli* with ceftriaxone, but only 0.824 on isolates from 2018, a drop of 0.057. Keeping the model useful therefore requires periodic retraining, and retraining requires labels. A label here is a susceptibility test with reagent and technician costs rather than an annotation that a person supplies in seconds, and a laboratory willing to fund only a few hundred extra tests in a year has to decide which isolates receive them.

This is the setting AL was designed for [3]. The arriving isolates are ranked by the model and labels are bought only where they are expected to be most informative, usually near the decision boundary [10] or spread across the feature space [11]. AL is well studied in medical imaging and in text, where it is routinely reported to reach a target accuracy with a fraction of the labels, but whether it does so for MALDI-TOF resistance prediction has not been established. A systematic review published in 2026, which covers 115 studies that apply ML to clinical MALDI-TOF spectra and groups them by task, preprocessing, model family and evaluation, does not identify a single study in which label acquisition cost is a variable [4]. Work on this modality either assumes that all labels are already present or evaluates a fixed training set.

We examine that gap directly. The experiments use DRIAMS, a public collection of clinical MALDI-TOF spectra with matched susceptibility results from 4 Swiss hospitals [2], released under a CC0 public domain waiver and archived at Dryad (doi:10.5061/dryad.bzkh1899q). The protocol is temporal rather than a random split. A model is fitted on isolates from 2015 and 2016, isolates arriving during 2017 are offered to an acquisition rule in 4 rounds under a fixed budget, and the maintained model is evaluated on isolates from 2018 (Fig. 1), which keeps the test year strictly after every label that was bought.

The contributions are as follows. This is the first evaluation of label acquisition cost for MALDI-TOF resistance prediction, covering 3 species and antibiotic pairs, 4 hospitals and 5 budgets, with 10 seeds per configuration. The main result is negative and holds across 135 paired comparisons,

979-8-3195-2818-6/26/$31.00 ©2026 IEEE

TABLE I
POSITION OF THIS STUDY RELATIVE TO REPRESENTATIVE PRIOR WORK. EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI) HERE MEANS ANY REPORTED FEATURE ATTRIBUTION.

Study	*Task and data*	*Sites*	*Temp.*	*Cost*	*Errors*	*XAI*
Weis *et al.* [2]	AMR from MALDI-TOF, DRIAMS	4	no	no	no	no
Weis *et al.* [5]	Topological and kernel methods	1	no	no	no	no
Duroux *et al.* [6]	Generalisation to unseen sites	4	yes	no	no	no
Corvelo Benz *et al.* [7]	Conformal guarantees, DRIAMS	4	no	no	partly	no
Schmidt-Santiago *et al.* [4]	Review of 115 MALDI-TOF ML studies	n/a	rare	none	rare	21/115
Chen *et al.* [8]	AL with selective prediction, images	n/a	no	yes	no	no
Chen *et al.* [9]	Continuous learning, malware	n/a	yes	yes	no	no
This work	AL for AMR from MALDI-TOF, DRIAMS	4	yes	yes	yes	yes

in that no acquisition rule outperformed random selection. We account for that result by measuring how little accuracy retraining makes available in the first place, which turns the useful quantity into cost, and we report how much of the benefit of retraining a partial test budget recovers (Table IV). Model quality next year and patient exposure this year are then separated, and the two disagree about which rule is preferable (Fig. 3). TreeSHAP finally shows that models trained under different acquisition rules rely on the same spectral regions (Fig. 4 and Table V), which is consistent with the absence of a difference between them.

II. RELATED WORK

A. Resistance prediction from mass spectra

Resistance prediction from MALDI-TOF spectra became a practical proposition with the release of DRIAMS, which pairs more than 300000 routine spectra with susceptibility results and established that gradient boosted trees and shallow networks reach clinically interesting discrimination on several species and antibiotic combinations [2]. The representation was later improved with topological and kernel methods [5], prediction was extended to several antibiotics at once, and statistical guarantees were added through conformal prediction [7]. SHAP values were used for interpretation in 21 of the 115 studies covered by the review of Schmidt-Santiago *et al.*, so model explanation is established practice on this modality, while temporal evaluation remains uncommon [4]. Generalisation across hospitals has been studied directly, and performance dropped sharply when a model was moved to a site it was not trained on [6].

B. Label acquisition under a budget

Active learning has a long history. Uncertainty sampling selects the instances closest to the decision boundary [10], core set and diversity methods instead cover the input space [11], and importance weighted and cost sensitive variants correct the bias that selective querying introduces [12], [13]. Hybrid rules combine the two signals [14]. The literature is not uniformly positive. Random selection is a stronger baseline than it is often given credit for, and injecting randomness into uncertainty sampling improved its robustness [15]. In security, where models are retrained continuously against a moving target, label acquisition under budget has been studied in its own right [9], [16], and work bridging AL and selective prediction shows that the two interact [8]. Those settings resemble ours in that labels are expensive and the data distribution moves, but none of them involves mass spectra or susceptibility testing.

C. Position of this study

Prior work on MALDI-TOF resistance prediction has not treated the acquisition of labels as a cost, has largely not used temporal splits, and has rarely reported the 2 clinical error types separately (Table I). Work on AL under budget, in turn, has not been applied to this modality. The combination of a temporal protocol, an explicit test budget, both error types, and an explanation of why the acquisition rules behave as they do is what this paper contributes.

III. METHOD

A. Problem setting

Let a spectrum be a vector x of 6000 binned intensities and let y be 1 if the isolate is resistant or intermediate to the antibiotic of interest and 0 if it is susceptible. A model f is fitted to the isolates available at the start of the period and outputs an estimated probability of resistance, and a threshold t turns that probability into a report (resistant when $f(x)$ is at least t, susceptible otherwise). Isolates then arrive over a year in 4 batches. The model scores every arriving isolate in each round, an acquisition rule selects $B/4$ of them for laboratory testing, where B is the budget for the whole year, and the true labels of those isolates are revealed and added to the training set before f and t are refitted. Isolates that were not selected are reported using the model's own output, which is the operational situation that matters clinically, because those reports reach clinicians without a confirmatory test (Fig. 1).

B. Acquisition rules

Four acquisition rules were compared. Random selection draws uniformly from the isolates that arrived in the round and serves as the baseline. Uncertainty sampling takes the isolates whose predicted probability is closest to the threshold t, measured as the absolute difference between the 2 [10]. Diversity sampling applies a k-center greedy procedure to the first 50 principal components of the spectra so that the selected isolates are spread out rather than clustered [11]. The hybrid rule takes the $4b$ isolates closest to the threshold, where b is the number bought in the round, groups them into b clusters by k-means on the same components, and buys the isolate nearest the centre of each cluster [14]. Two bounds frame all four. The lower bound never retrains at all, and the upper bound retrains

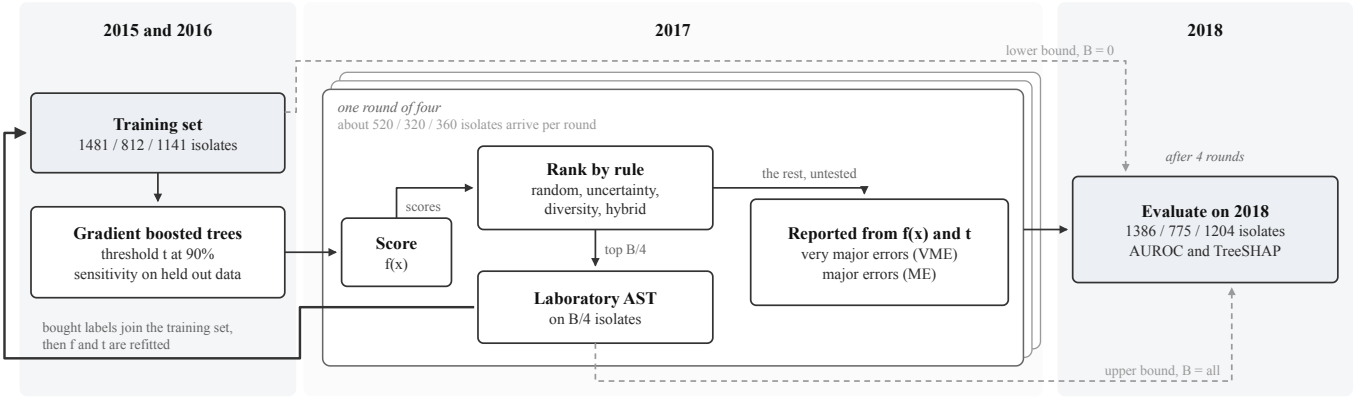


Fig. 1. Study pipeline. The 3 counts in each box are the isolates available for *E. coli*, *K. pneumoniae* and *S. aureus* at DRIAMS-A. One of the 4 rounds of 2017 is drawn, in which a quarter of the annual budget B is spent on laboratory susceptibility testing (AST) and those labels enter the training set. Isolates that are not bought are reported from the model output alone, and the dashed arms are the 2 bounds.

on every isolate that arrived, which is what a laboratory would obtain if it tested everything.

C. Error types

Clinical microbiology distinguishes two kinds of mistake, and they are not interchangeable. A very major error (VME) is a resistant isolate reported as susceptible, which can send a patient home on a drug that will not work. A major error (ME) is a susceptible isolate reported as resistant, which wastes a broader spectrum agent and contributes to selection pressure. The definitions follow the International Organization for Standardization (ISO) standard 20776-2 [17]. Both are reported throughout, because a rule that lowers one while raising the other has not improved anything.

D. Predictive and operational views

Two views that are often conflated are kept apart here. The predictive view is the usual one, namely how well next year's model discriminates, measured as AUROC on the 2018 isolates. The operational view counts what happened during the pool year itself, namely how many of the isolates that were not selected for testing were resistant and were nonetheless reported susceptible. These are counts of patients affected rather than rates, and the two views do not have to agree.

E. Explaining the fitted models

SHAP values were computed to understand why the rules behave similarly [18]. They distribute a single prediction across the input features in the unique way that satisfies local accuracy, missingness and consistency, and for tree ensembles they have an exact polynomial time solution, so no sampling approximation is involved. We used the TreeSHAP implementation inside the gradient boosting library itself [19], which removes any possibility of a mismatch between the explained model and the deployed one. Six models were explained per pair (the two bounds, and random and uncertainty acquisition at budgets of 400 and 800 labels) on the same 2018 isolates, so any difference is a difference between models rather than between evaluation sets. Two models were then compared by the overlap of their 20 most used bins and by the rank correlation over all 6000 bins. Importance of a bin is its mean absolute SHAP value, and its direction was obtained by splitting the isolates at the median intensity of that bin and taking the difference in mean SHAP between the halves, because averaging signed values over all isolates simply follows the majority class (here susceptible).

IV. Experimental Setup

A. Data and splits

DRIAMS provides spectra preprocessed to 6000 bins of 3 daltons each, spanning 2000 to 20000 daltons, and labels follow the convention of the original release, in which resistant and intermediate are coded together as 1. Three species and antibiotic pairs have enough labelled isolates for a temporal analysis and were studied here, namely *E. coli* with ceftriaxone, *Klebsiella pneumoniae* with ceftriaxone, and *Staphylococcus aureus* with oxacillin. At the primary site (DRIAMS-A) these gave 1481, 812 and 1141 training isolates from 2015 and 2016, pools of 2094, 1273 and 1445 isolates arriving in 2017, and test sets of 1386, 775 and 1204 isolates from 2018, with resistance prevalence in the test year of 18.5%, 13.9% and 14.6% respectively. The released spectra carry no acquisition dates, so the 4 arrival batches were formed by a deterministic hash of the isolate code rather than by calendar quarter (the MD5 digest of the code modulo 4, which is the same for every rule and seed and gives batches of about a quarter of the pool each). The batches decide which isolates compete for the budget of a round, but their order carries no calendar information.

B. Cross site protocol

The cross site experiment moves a model trained at DRIAMS-A to each of DRIAMS-B, DRIAMS-C and DRIAMS-D. The source model was fitted on the DRIAMS-A isolates of all 4 years (4961, 2860 and 3790 for the 3 pairs,

TABLE II

CROSS SITE EXPERIMENT. POOL AND TEST ARE THE 2 HALVES OF THE 2018 ISOLATES OF THE TARGET SITE, WITH THE RESISTANT ISOLATES OF THE TEST HALF IN PARENTHESES. NONE AND ALL ARE THE TEST AUROC OF THE SOURCE MODEL USED UNCHANGED AND OF THE MODEL REFITTED ON THE WHOLE POOL. THE LAST 3 COLUMNS ARE THE MEAN PAIRED AUROC DIFFERENCE OF EACH RULE AGAINST RANDOM SELECTION OVER THE BUDGETS SMALLER THAN THE POOL, AND A DAGGER MARKS A 95% CONFIDENCE INTERVAL THAT EXCLUDES ZERO.

Pair	*Site*	*Pool*	*Test*	*None*	*All*	*Unc.*	*Div.*	*Hyb.*
E. coli	B	107	106 (22)	0.835	0.860	−0.007	+0.002	−0.007
	C	458	458 (76)	0.659	0.705	−0.006	+0.001	−0.006
	D	997	997 (99)	0.777	0.788	−0.002	−0.004	−0.000
K. pneu.	B	76	76 (9)	0.477	0.614	−0.020	−0.059†	−0.049†
	C	182	184 (28)	0.585	0.682	−0.031†	−0.014	−0.007
	D	1075	1076 (62)	0.616	0.638	−0.004	−0.007	−0.003
S. aureus	B	174	172 (10)	0.800	0.829	+0.000	−0.006	−0.013
	C	370	368 (20)	0.648	0.691	+0.010	+0.006	−0.001

a fifth of them held out for the threshold). The 2018 isolates of the target site were divided in half, stratified by label and redrawn for every seed, into a pool from which labels could be bought and a test set (Table II). The budget (25, 50, 100, 200 or 400 labels) was spent in a single round, the model was refitted on the source isolates together with the bought ones at equal weight, and the 2 bounds are the source model used unchanged and the model refitted on the whole pool. *S. aureus* with oxacillin has no labelled isolates at DRIAMS-D, which leaves 8 combinations of pair and target site.

C. Model and protocol

The classifier is a gradient boosted tree ensemble [19], tuned once on the 2015 and 2016 data only and then held fixed, so that no information from the pool or test years entered the configuration. Of the training years, 20% was held out to set the threshold t at the highest value that still gave 90% sensitivity to resistance, which reflects the clinical preference for avoiding very major errors. Budgets of 40, 100, 200, 400 and 800 labels were spread evenly over the 4 rounds. Every configuration was repeated with 10 random seeds, and the cross site configurations with 5, giving 1540 retraining simulations in total (660 temporal, from 3 pairs, 22 configurations made of 4 rules at 5 budgets and the 2 bounds, and 10 seeds, and 880 cross site, from 8 combinations of pair and site, 22 configurations and 5 seeds). A seed fixes the held out fifth of the training data, the draws of random selection, the subsampling inside the model and, in the cross site setting, the division of the target site into pool and test set, so a rule and random selection at the same seed differ only in which isolates were bought.

D. Statistics and availability

Each rule was compared with random selection on matched seeds, the paired quantity being the difference in test AUROC between the rule and random selection at the same pair, site, budget and seed. The test is the exact two sided Wilcoxon signed rank test, with 10 pairs of runs per comparison in the temporal setting and 5 in the cross site setting, and the resulting p values were corrected with the Holm procedure [20] within each species and antibiotic pair (15 comparisons, from 3 rules and 5 budgets) and also across all 45 temporal comparisons. An exact test on 10 seeds cannot return a value below 0.002, so no temporal comparison can fall below 0.029 after correction within a pair or below 0.088 after correction across all 45, and on 5 seeds the floor is 0.0625 before any correction, which means that the cross site tests could not have reached 0.05 whatever the data. The effect size is therefore given for every comparison as the mean paired difference in AUROC with its 95% confidence interval (Student t, 9 or 4 degrees of freedom). The DRIAMS release used here is archived at doi:10.5061/dryad.bzkh1899q, and our code, the per isolate acquisition log and the per run result records are released with the paper.

V. RESULTS

A. Bounds on the available gain

Two bounds define the space in which any acquisition rule has to operate. A model that was never retrained reached an AUROC on the 2018 isolates of 0.823, 0.742 and 0.799 for the 3 pairs, while a model retrained on every isolate that arrived in 2017 reached 0.867, 0.788 and 0.850. Retraining helps, and the direction of the effect was consistent across all 3 pairs, which confirms the premise of the study. The size of the effect is modest. The difference between the 2 bounds was 0.044 (*E. coli*), 0.046 (*K. pneumoniae*) and 0.051 (*S. aureus*) of AUROC, so every acquisition rule competes inside a window of about 5 points, and the standard error over 10 seeds was of a similar order at the smaller budgets (Fig. 2).

B. Comparison against random selection

All 4 rules improved with budget and all of them stayed between the 2 bounds, but they were not separable from one another (Fig. 2). The paired tests make this precise. The temporal setting gives 45 comparisons against random selection (3 pairs, 5 budgets and 3 competing rules), in which the mean paired difference ran from 0.029 AUROC below random selection to 0.012 above it. Before correction 14 of them reached $p < 0.05$, 12 in favour of random selection and 2 in favour of the rule (diversity sampling on *K. pneumoniae* at 200 labels and the hybrid rule on *E. coli* at 800, each better by about 0.01). Corrected within each pair, 2 remained below 0.05 and both favoured random selection. Uncertainty sampling on *K. pneumoniae* at 800 labels was worse by 0.029 AUROC (95% confidence interval 0.014 to 0.045) and diversity sampling on *S. aureus* at 40 labels was worse by 0.008 (0.003 to 0.013), each at a corrected p of 0.029, which is the floor of the design. Corrected across all 45 the smallest value is 0.088, so under the stricter family no comparison is significant in either direction, and the 2 losses are the largest effects observed rather than established ones. Averaged over the 5 budgets, no rule had an interval above zero for any pair, and uncertainty sampling on *K. pneumoniae* was worse by 0.017 (0.010 to 0.025).

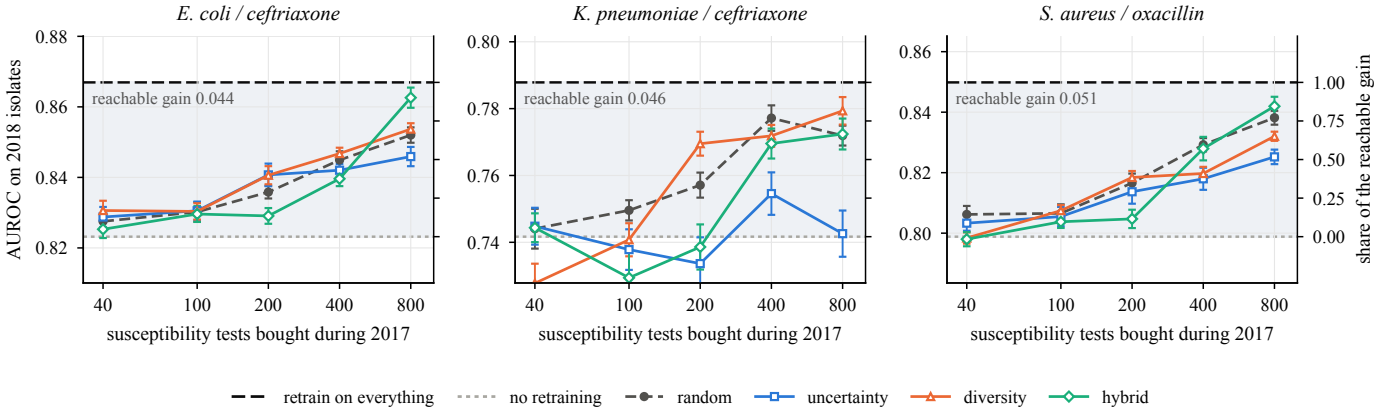


Fig. 2. Test AUROC on 2018 isolates against the number of susceptibility tests bought during 2017, for each species and antibiotic pair. Points are means over 10 seeds and bars are standard errors. The shaded band runs from never retraining (dotted line) to retraining on every arriving isolate (dashed line), and its height is printed inside each panel. The right hand axis reads the same band as a share of the gain retraining makes available. All 4 rules lie inside the band and their intervals overlap at almost every budget.

The cross site experiment adds 120 further comparisons across DRIAMS-B, DRIAMS-C and DRIAMS-D. In 30 of them the budget was at least as large as the pool of the target site, so every rule bought the same isolates and the difference is zero by construction, which leaves 90 that are informative. None of these can reach significance on 5 seeds, so they are read as effect sizes. The mean difference was negative in 64 of the 90 and positive in 26, it ran from 0.073 below random selection to 0.019 above it, and its 95% interval excluded zero in 7 comparisons (6 against the rule and 1 for it), against the 4 or 5 that chance would produce at that level. Averaged over the informative budgets, 3 of the 24 combinations of pair, site and rule had an interval below zero and none had one above it (Table II). These 135 paired comparisons produce no evidence that an informed rule helps, and where a difference can be detected it favours random selection. The result held over 3 species, 2 antibiotic classes, 4 hospitals and 2 deployment settings, including the cross site case in which the new data differs most from the training data and in which AL would be expected to help most.

C. Sensitivity to the arrival batches

Because the batches are a pseudo ordering, the comparison was repeated with the same data, model and seeds but with the 2017 pool divided differently (Table III). Three further divisions were drawn from independent hashes of the isolate code, and in a fourth setting the sequence was removed altogether and the whole budget was bought in a single round. Budgets of 100, 400 and 800 labels were used, which gives 27 comparisons per division, each corrected within its pair (9 comparisons). The division of the main experiment was rerun alongside as a reference, and it reproduced all 360 of the corresponding runs exactly. The picture did not change. The mean difference was negative in 16 to 20 of the 27 comparisons of every division, and after correction 8 comparisons across the 5 divisions favoured random selection against 1 that favoured a rule (diversity sampling on *K. pneumoniae* at 800 labels in one redrawn division, better by 0.025, 95% confidence interval 0.014 to 0.036). That comparison was positive in all 5 divisions, by 0.004 to 0.025, and it is the only advantage of any rule that recurred. Uncertainty sampling on *K. pneumoniae* at 800 labels was worse in every division, by 0.021 to 0.062. The operational result was equally stable. Uncertainty sampling at 800 labels reduced the resistant isolates reported susceptible in every division and for every pair (by 33% to 43% for *E. coli*, 52% to 73% for *K. pneumoniae* and 51% to 67% for *S. aureus*), with a 95% interval that excluded zero in all 15 cases.

TABLE III

THE TEMPORAL COMPARISON UNDER OTHER DIVISIONS OF THE 2017 POOL INTO ARRIVAL BATCHES (27 COMPARISONS PER DIVISION, BUDGETS OF 100, 400 AND 800 LABELS). MIN AND MAX ARE THE SMALLEST AND LARGEST MEAN PAIRED AUROC DIFFERENCE OF A RULE AGAINST RANDOM SELECTION, NEG. COUNTS THE NEGATIVE ONES, AND AGAINST AND FOR COUNT THE COMPARISONS BELOW 0.05 AFTER CORRECTION WITHIN EACH PAIR. THE LAST 3 COLUMNS ARE THE CHANGE IN RESISTANT ISOLATES REPORTED SUSCEPTIBLE UNDER UNCERTAINTY SAMPLING AT 800 LABELS.

Division	*Min*	*Max*	*Neg.*	*Against*	*For*	*E. c.*	*K. p.*	*S. a.*
Main experiment	−0.029	+0.011	18	1	0	−33%	−73%	−51%
Redrawn 1	−0.024	+0.025	16	1	1	−35%	−58%	−51%
Redrawn 2	−0.044	+0.011	19	1	0	−34%	−68%	−67%
Redrawn 3	−0.021	+0.009	20	2	0	−43%	−52%	−65%
Single round	−0.062	+0.009	17	3	0	−33%	−60%	−53%

D. Cost of a partial test budget

If the rules are interchangeable, the question worth asking is how many tests to buy rather than which rule to use. Testing every isolate that arrives in a year costs 2094, 1273 and 1445 susceptibility tests at DRIAMS-A, and a budget of 400 tests (19% to 31% of those totals) recovered about half of the available gain for *E. coli* (0.50) and roughly three quarters for *K. pneumoniae* (0.77), which is what Table IV expresses as a price. The curve of benefit against labels is steep at the start and flat later (Fig. 2), so most of the value lies in the decision

TABLE IV

SHARE OF THE AVAILABLE AUROC GAIN RECOVERED AT EACH BUDGET, AVERAGED OVER 10 SEEDS. A VALUE OF 1.00 WOULD MATCH RETRAINING ON EVERY ISOLATE AND 0.00 WOULD MATCH NEVER RETRAINING. THE ANTIBIOTIC IS CEFTRIAXONE FOR THE FIRST 2 PAIRS AND OXACILLIN FOR THE THIRD.

Pair	*Tests bought*	*Of pool*	*Tests saved*	*Random*	*Uncert.*	*Divers.*	*Hybrid*
E. coli	400	2094	80.9%	0.50	0.43	0.54	0.38
	800	2094	61.8%	0.66	0.52	0.70	0.90
K. pneumoniae	400	1273	68.6%	0.77	0.28	0.65	0.60
	800	1273	37.2%	0.66	0.02	0.82	0.67
S. aureus	400	1445	72.3%	0.60	0.38	0.41	0.57
	800	1445	44.6%	0.77	0.52	0.65	0.85

TABLE V

AGREEMENT BETWEEN FITTED MODELS IN THEIR SHAP PROFILES. J IS THE JACCARD OVERLAP OF THE 20 MOST USED SPECTRAL BINS, WHERE 1.00 MEANS IDENTICAL LISTS AND 0.00 MEANS NO BIN IN COMMON. ρ IS THE SPEARMAN RANK CORRELATION OF IMPORTANCE OVER ALL 6000 BINS.

	E. coli		*K. pneu.*		*S. aureus*	
Models compared	*J*	ρ	*J*	ρ	*J*	ρ
Random 400 vs uncertainty 400	0.60	0.38	0.38	0.44	0.54	0.44
Random 800 vs uncertainty 800	0.60	0.39	0.29	0.44	0.43	0.48
Random 800 vs retrain on all	0.67	0.40	0.38	0.52	0.60	0.54
Uncertainty 800 vs retrain on all	0.60	0.39	0.38	0.48	0.43	0.47
No retraining vs retrain on all	0.43	0.24	0.11	0.22	0.29	0.30
Mean over all 15 model pairs	0.60	0.36	0.31	0.40	0.49	0.44

to retrain at all, and comparatively little in the decision about which isolates to retrain on.

E. Operational errors during the pool year

AUROC describes next year's model and says nothing about the patients whose isolates were reported during the year in which the labels were being collected, which is what Fig. 3 counts. At a budget of 800 tests the number of resistant isolates that went untested and would have been reported susceptible fell from 49.8 to 33.6 for *E. coli* under uncertainty sampling rather than random selection, from 19.4 to 5.2 for *K. pneumoniae*, and from 18.8 to 9.2 for *S. aureus*, reductions of 33%, 73% and 51%. Major errors moved in the same direction for 2 of the 3 pairs, falling from 347.8 to 259.8 for *E. coli* and staying level for *S. aureus* (134.6 to 140.0), but they rose from 123.7 to 190.1 for *K. pneumoniae*, which is the one configuration in which the safety gain is paid for in broad spectrum prescribing.

The mechanism is visible in the acquisition log, which records one line per isolate per round. Contrary to what might be expected, uncertainty sampling bought fewer resistant isolates than random selection did. The resistant share of what it bought at a budget of 400 labels was 14.9%, 9.7% and 10.8%, against arrival rates of 22.2%, 14.2% and 18.8%. An isolate that the model confidently calls resistant lies far from the threshold, so the rule declines to pay for it, and the isolates it does buy sit 2 to 4 times closer to the threshold (median absolute distance 0.008 against 0.030 for *E. coli*) than those it skips. Uncertainty sampling is therefore a detector of impending mistakes rather than a detector of resistance. It finds the cases the model was about to get wrong, and it does so while spending less of the budget on cases the model already handles correctly.

F. SHAP profiles of the fitted models

Models that differ only in how their labels were chosen agreed with each other about as much as either agreed with the model retrained on everything (Table V). The comparison that stands out is the last one before the mean. The model that was never retrained was the outlier in every pair, and on *K. pneumoniae* it shared only 2 of 20 bins with the fully retrained model (Jaccard index 0.11, ρ = 0.22). Retraining changes what a model looks at, whereas changing the acquisition rule at a fixed amount of retraining largely does not.

Counting how many of the 6 models placed a given bin in their top 20 makes the structure clearer (Fig. 4). Nine bins appeared in all 6 lists for *E. coli*, out of 32 that appeared anywhere, and the strongest lay at 2150 to 2156 daltons. The figure was 8 of 40 for *S. aureus*, at 2405 to 2411 and 2852 to 2858 daltons, and only 2 of 57 for *K. pneumoniae*, which matches that pair having the fewest isolates, the lowest AUROC and the least stable explanations. A small shared core is found by every model and the tail shifts with the training set. Because all 6 models are fitted to the same underlying relationship, and the easily located part of it is located by all of them, similar accuracy is the expected outcome. We do not assign these masses to particular proteins. The low mass region is crowded, several known markers fall within a few daltons of one another, and a 3 dalton bin cannot separate them.

VI. DISCUSSION

A. Why label efficiency is the wrong objective here

The natural reading of a negative result is that the method was implemented badly. The better reading here is that label efficiency is the wrong objective for this problem. AL promises to reach a given accuracy with fewer labels, and that promise is only valuable when the accuracy at stake is large. On this modality the entire gap between never retraining and retraining on everything is around 5 points of AUROC (0.044 to 0.051), and the training years already contain thousands of examples of the few spectral peaks that carry the signal. One more year of isolates adds mostly repetition, and choosing which repetitions to buy is a small lever on a problem whose bottleneck lies elsewhere. The SHAP analysis supports this reading, since the models converged on the same bins regardless of how their labels were selected (Fig. 4).

B. What a laboratory should take from this

The acquisition rule still matters, although for a different reason. The rules differ in which patients are exposed to an error while the labels are being gathered, and that difference is large where the accuracy difference is small. A laboratory choosing a policy has to decide whether it is optimising next

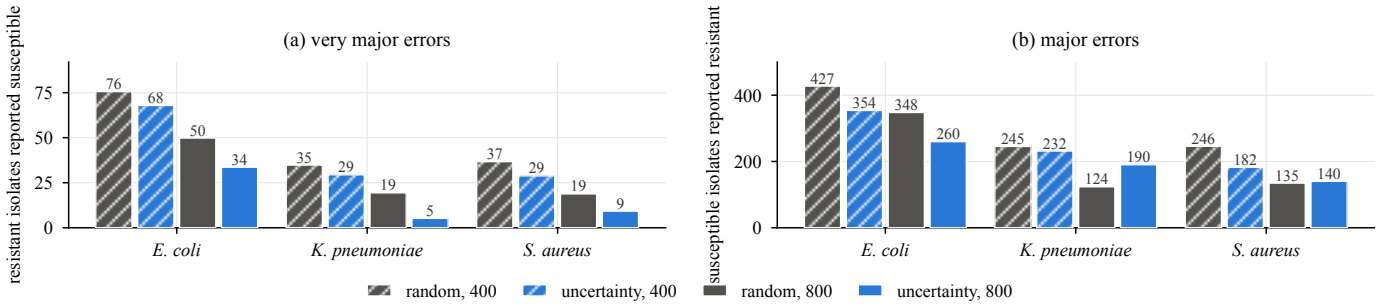


Fig. 3. Clinical errors on the isolates that were not selected for testing, summed over the pool year and averaged over 10 seeds. Hatched bars are a budget of 400 tests and solid bars 800. Panel (a) counts resistant isolates that would have been reported susceptible, panel (b) susceptible isolates that would have been reported resistant.

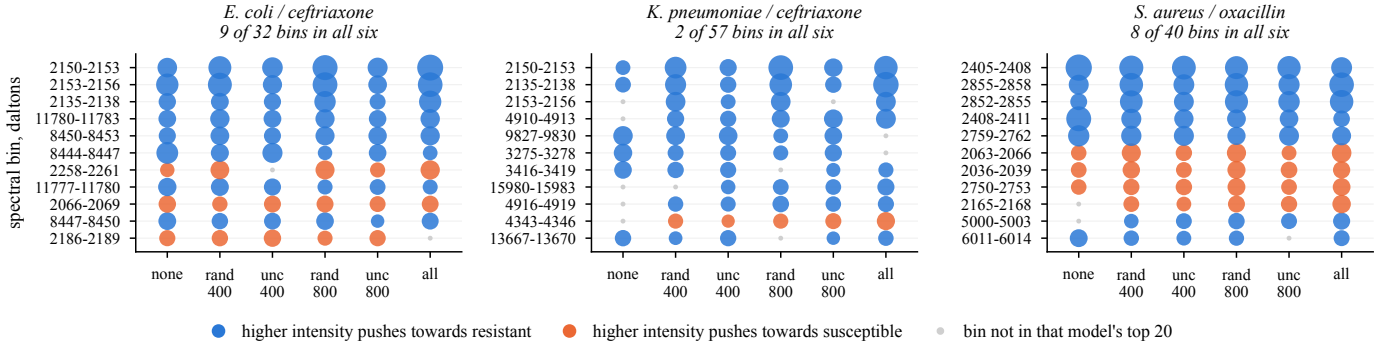


Fig. 4. Spectral bins the 6 explained models rely on, for the 11 most important bins that reached the top 20 of at least 4 of them. Marker area is the mean absolute SHAP value of that bin for that model, colour is the direction of the effect, and grey dots mark bins outside that model's top 20. Columns are the 2 bounds (none, all) and random and uncertainty acquisition at 400 and 800 labels.

year's model or this year's reports. A budget of a few hundred tests selected by uncertainty is a reasonable default on these data, since it costs the least, it gives up little AUROC against random selection (0.017 on average over the budgets for *K. pneumoniae*, 0.006 for *S. aureus* and nothing measurable for *E. coli*), and it substantially reduces the resistant isolates that would otherwise be reported susceptible. The *K. pneumoniae* result at 800 labels is the caveat, both because it is the one configuration where uncertainty sampling lost AUROC outright and because it is the one where major errors rose (Fig. 3), and a laboratory working on that organism would be buying a 73% reduction in very major errors at the price of 66 additional major errors over the year.

A major error puts a patient on a broader agent than was needed until the culture based result arrives 1 to 3 days later, and it adds to the selection pressure that stewardship tries to limit, whereas a very major error leaves an infection on a drug that does not work. About 5 additional major errors for each very major error avoided (66 against 14) is a trade that a laboratory may accept, but it is a trade rather than a free gain. Its origin is visible in the first round, in which both rules face the same model and the same threshold. With t as low as 0.014 the isolates nearest to it are those the model would report susceptible, so a budget that covers 63% of the pool (800 of 1273) confirms nearly all of the susceptible reports by testing and leaves the resistant reports unconfirmed, and 75% of the untested susceptible isolates then lay above the threshold, against 42% under random selection. The model handed to the following year shows a related shift. On the 2018 isolates the models maintained by uncertainty sampling made more very major errors and fewer major errors than those maintained by random selection in 14 of the 15 configurations (at 400 labels for *E. coli*, rates of 0.22 against 0.14 and of 0.29 against 0.45), so the rule moves the operating point as well as the training set, and a laboratory that adopts it should set t again on recent labels rather than keep the value fitted to the training years.

C. Limitations

The released spectra carry no acquisition dates, so the arrival batches within 2017 are a deterministic pseudo ordering rather than true calendar quarters, although the year boundaries themselves are real. The simulation therefore represents drift between years but not within a year, and anything that changes during a year (the seasonal mix of isolates, an outbreak, a serviced or replaced instrument) reaches every batch in equal measure. A rule that adapts to recent isolates could gain from a real ordering in a way these experiments cannot show, and redrawing the batches establishes only that the result does not depend on one particular division. The 3 pairs studied are those with enough labelled isolates for a temporal analysis, and the conclusions may not transfer to rarer organisms or

to antibiotics with very low resistance prevalence. *S. aureus* with oxacillin has almost no resistant isolates at 2 of the 4 hospitals, which limits the power of the cross site comparisons for that pair. The explanations were computed from 1 seed per configuration, so some of the disagreement in the tail of the ranking is sampling noise. All of this is a retrospective simulation on stored spectra from the hospitals of a single country, in which a bought label is available at once, and a prospective study would have to contend with laboratory turnaround (the result of a susceptibility test returns after the report has been issued), with clinicians who may override the model, and with instrument changes that our protocol holds fixed. The operational counts are what the reports would have been rather than what happened to patients, so the safety result in particular needs confirmation on isolates collected in real time, with their dates, before it is used to set a testing policy.

VII. Conclusion

Label acquisition rules for the maintenance of MALDI-TOF resistance prediction models were evaluated under a temporal protocol on DRIAMS. Across 135 paired comparisons spanning 3 species and antibiotic pairs, 4 hospitals and 5 budgets, no uncertainty, diversity or hybrid rule outperformed random selection, and the differences that could be detected favoured random selection. The same held when the pool year was divided into batches in 4 other ways. The explanation is that retraining makes only 0.044 to 0.051 of AUROC available in total, and that models trained under different rules rely on the same small set of spectral bins (Table V). The quantity worth reporting is therefore cost rather than label efficiency, in that roughly a third of a year of susceptibility tests recovers between half and three quarters of the benefit of testing everything (Table IV). Uncertainty sampling remains worth using for a different reason, since it reduced by a third to three quarters the resistant isolates that were reported susceptible while the labels were being collected. Future work should test whether these conclusions hold prospectively, on isolates with real acquisition dates, and whether acquisition rules that target the clinical error directly, rather than model uncertainty, do better on the operational view.

Acknowledgment

An AI assistant (Claude, Anthropic) was used to support the writing of the analysis code and the recomputation of the statistical tests. The authors take full responsibility for the content of the paper.

References

[1] Antimicrobial Resistance Collaborators, "Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis," *The Lancet*, vol. 399, no. 10325, pp. 629-655, 2022.

[2] C. Weis *et al.*, "Direct antimicrobial resistance prediction from clinical MALDI-TOF mass spectra using machine learning," *Nature Medicine*, vol. 28, pp. 164-174, 2022.

[3] B. Settles, "Active learning literature survey," Computer Sciences Technical Report 1648, University of Wisconsin-Madison, 2009.

[4] L. Schmidt-Santiago, A. Guerrero-Lopez, C. Sevilla-Salcedo, D. Rodriguez-Temporal, B. Rodriguez-Sanchez, and V. Gomez-Verdejo, "A systematic review of machine learning on clinical MALDI-TOF MS," *Briefings in Bioinformatics*, vol. 27, 2026, doi:10.1093/bib/bbag208.

[5] C. Weis, M. Horn, B. Rieck, A. Cuenod, A. Egli, and K. Borgwardt, "Topological and kernel-based microbial phenotype prediction from MALDI-TOF mass spectra," *Bioinformatics*, vol. 36, no. S1, pp. i30-i38, 2020.

[6] D. Duroux, P. P. Meyer, G. Visona, and N. Beerenwinkel, "Generalizable machine learning models for rapid antimicrobial resistance prediction in unseen health care settings," *GigaScience*, 2026, doi:10.1093/gigascience/giaf156.

[7] N. Corvelo Benz, L. Miranda, D. Chen, J. Sattler, and K. Borgwardt, "Detecting antimicrobial resistance through MALDI-TOF mass spectrometry with statistical guarantees using conformal prediction," in *Research in Computational Molecular Biology (RECOMB)*, Springer, 2025, doi:10.1007/978-3-031-90252-9_5.

[8] J. Chen, J. Yoon, S. Ebrahimi, S. O. Arik, S. Jha, and T. Pfister, "ASPEST: bridging the gap between active learning and selective prediction," *Transactions on Machine Learning Research*, 2024.

[9] Y. Chen, Z. Ding, and D. Wagner, "Continuous learning for Android malware detection," in *USENIX Security Symposium*, 2023.

[10] D. D. Lewis and W. A. Gale, "A sequential algorithm for training text classifiers," in *ACM SIGIR*, 1994, pp. 3-12.

[11] O. Sener and S. Savarese, "Active learning for convolutional neural networks: a core set approach," in *International Conference on Learning Representations (ICLR)*, 2018.

[12] A. Beygelzimer, S. Dasgupta, and J. Langford, "Importance weighted active learning," in *International Conference on Machine Learning (ICML)*, 2009, pp. 49-56.

[13] A. Krishnamurthy, A. Agarwal, T.-K. Huang, H. Daumé III, and J. Langford, "Active learning for cost-sensitive classification," *Journal of Machine Learning Research*, vol. 20, no. 65, pp. 1-50, 2019.

[14] W.-N. Hsu and H.-T. Lin, "Active learning by learning," in *AAAI Conference on Artificial Intelligence*, 2015, pp. 2659-2665.

[15] P.-Y. Lu, C.-L. Li, and H.-T. Lin, "A more robust baseline for active learning by injecting randomness to uncertainty sampling," in *AI and HCI Workshop at ICML*, 2023.

[16] F. Barbero, F. Pendlebury, F. Pierazzi, and L. Cavallaro, "Transcending Transcend: revisiting malware classification in the presence of concept drift," in *IEEE Symposium on Security and Privacy*, 2022, pp. 805-823.

[17] International Organization for Standardization, *ISO 20776-2:2021, Susceptibility testing of infectious agents and evaluation of performance of antimicrobial susceptibility test devices*, 2021.

[18] S. M. Lundberg *et al.*, "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, pp. 56-67, 2020.

[19] T. Chen and C. Guestrin, "XGBoost: a scalable tree boosting system," in *ACM SIGKDD*, 2016, pp. 785-794.

[20] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, pp. 65-70, 1979.